COSC 236 GRASP Report

Information Expert

The Information Expert Principle was applied to the Book, Member, and Library classes.

The Book class knows its own availability, updating its own status using borrow() and returnBook().

The Member class knows what books it has borrowed, keeping track of a list of borrowed books using borrowBook() and returnBook().

The Library class knows what members have registered and what books are catalogued, retrieving information from both lists using findBookByTitle() and findMember().

Creator

The Library class manages the catalogue of books and the list of registered members, adding and removing objects from both lists using addBook(), removeBook(), registerMember(), and removeMember().

Controller

The LibrarianController class handles requests using the borrowBook() and returnBook() methods, delegating both tasks to the Book and Member classes.

Low Coupling & High Cohesion

Each class interacts with the others through public methods instead of directly modifying their attributes, promoting Low Coupling. Each class also works on one responsibility (Book tracks its availability, Member tracks borrowed books, Library tracks what books/members are part of the system, LibrarianController handles requests), promoting High Cohesion